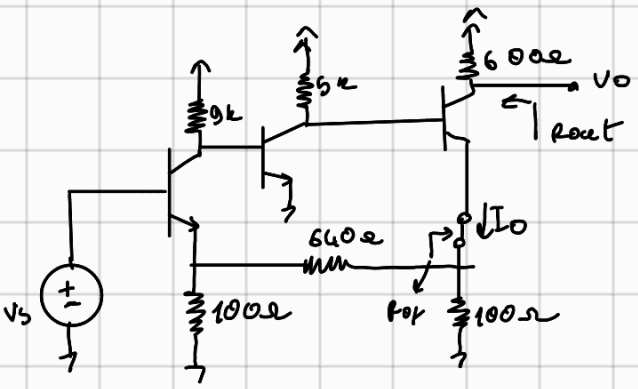
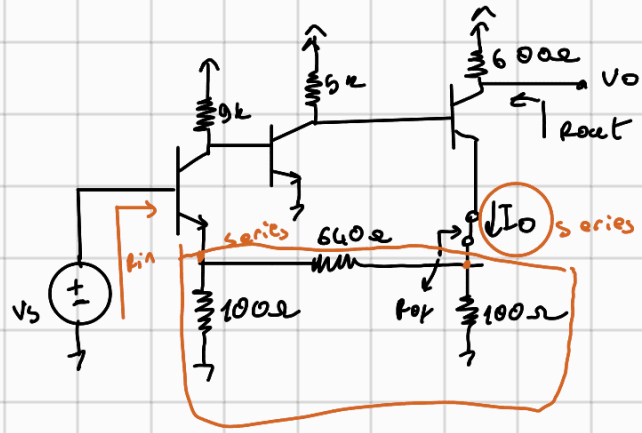
Analyze the circuit given below (Transconductance Amplifier) (Series – Series)

$I_{C1} = 0,6mA$, $I_{C2} = 1mA$, $I_3 = 4mA$

$[\beta = 100, r_o = \infty]$ Find $A, \beta, A_f = \frac{I_o}{V_s}, \frac{V_o}{V_s}$





$g_{m1} = \frac{0,6mA}{25mV} = 24mS$

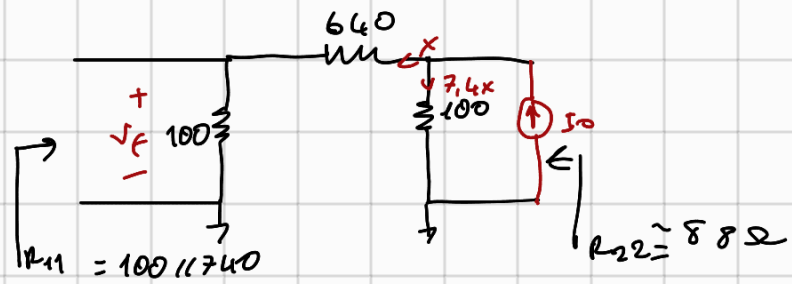
$g_{m2} = \frac{1mA}{25mV} = 40mS$

$g_{m3} = \frac{4mA}{25mV} = 10mS$

$r_{\pi 1} = \frac{1000}{24} \times 100 = 4,2k$

$r_{\pi 2} = 2,5k$

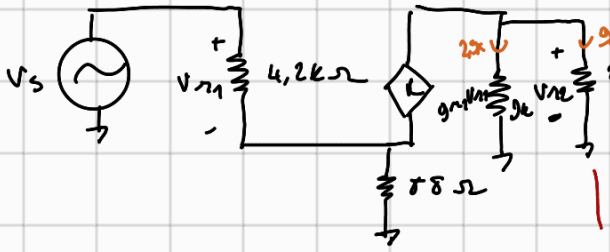
$r_{\pi 3} = 0,625k$

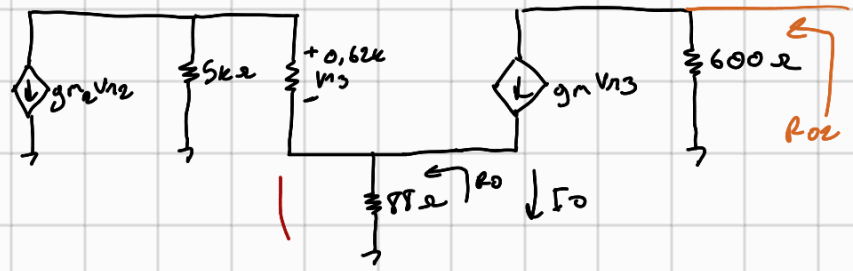


$= 100 \| 7440$

$= 88\Omega$

$\beta = \frac{V_f}{I_o} = \frac{100x}{8,4x} = 11,9\Omega$





$$V_s = V_{\pi 1} + 88\left(0{,}024\,V_{\pi 1} + \frac{V_{\pi 1}}{4200}\right)$$

$$V_s \cong 3{,}03\,V_{\pi 1}$$

$$V_{\pi 2} = \frac{-24\ mS \cdot V_{\pi 1} \times 2{,}5k \times 9k}{11{,}5k}$$

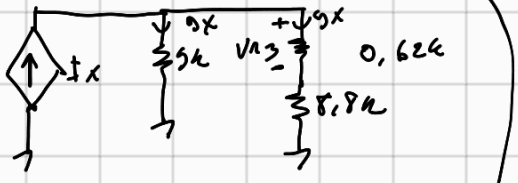
$$V_{\pi 2} = -47\,V_{\pi 1}$$

$$I_o = g_{m3}V_{\pi 3} + \frac{V_{\pi 3}}{0{,}62k}$$

$(R_o \cdot \beta + 1 + 0{,}62k)$

$$V_{\pi 3} = 3{,}1\times = -8{,}56\,V_{\pi 2}$$

$$= -g_{m2}\frac{V_{\pi 1} \cdot 3{,}1}{14}$$



$$I_x = -g_m\,V_\pi$$

$$\frac{I_x \cdot 9k}{(14k)} \cdot 0{,}62k$$

$$V_s = \frac{-3{,}03}{47} \cdot \frac{-I_o}{8{,}57} \cdot \frac{1}{2{,}16} = \frac{I_o}{V_o} \cong 21{,}27\,[A/V]$$

$$R_i = 4{,}2k + 88 \times 100$$

$$R_i = 4{,}2k + 8{,}8k = 13k\Omega$$

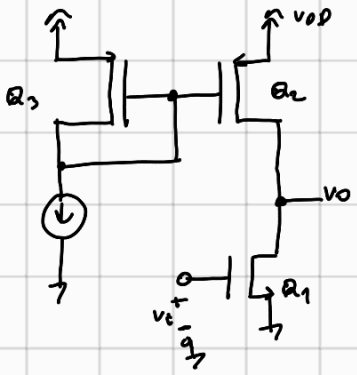
$$R_o = 88 + \frac{5k + 0{,}62k}{100} = 144{,}2\Omega$$

$$R_{o2} = 600\Omega$$

$$R_{iF} = (1 + A\beta)R_i = 1085\Omega$$

$$R_{oF} = (1 + A\beta)R_o = 12{,}09\Omega$$

$$A_F = \frac{21{,}27}{1 + 21{,}27 \times 11{,}9} = 0{,}0837$$



Consider the active loaded common source amplifier

given at left $\left(\frac{W}{L}\right) = \frac{8\,\mu m}{0{,}4\,\mu m}$ for all trans

$\mu_n C_{ox} = 387\,\mu A/V^2$

$\mu_p C_{ox} = 86\,\mu A/V^2$

$I_{ref} = 100\,\mu A$

$V_A'n = 5\,V/\mu m$

$V_A'p = 10\,V/\mu m$

$V_A = V_A' L$

For Q_1, $C_{gs} = 20\,fF$, $C_{gd} = 5\,fF$, $C_L = 25\,fF$ and $R_{sig} = 10\,k\Omega$. Assuming that C_L represents all the capacitance at that node

Find f_H with open time constant and Miller approaches

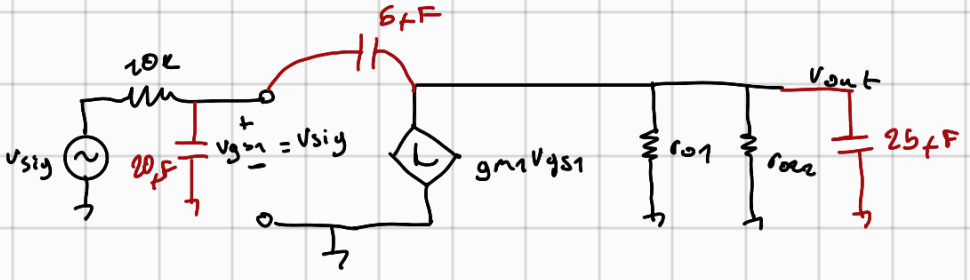
$$I_D = 100\,\mu A = \frac{1}{2} 387\,\mu A/V^2 \cdot 20\,(V_{OV})^2$$

$$V_{OV} = \sqrt{\frac{10}{387}} = \pm 0{,}16 \rightarrow +0{,}16\,V$$

$$g_{m1} = 2 \frac{100\,\mu A}{0{,}16\,V} = 1{,}25\,mS$$

$$r_{o1} = \frac{5\,V/\mu m \times 0{,}4\,\mu m}{100\,\mu A} = 20\,k\Omega$$

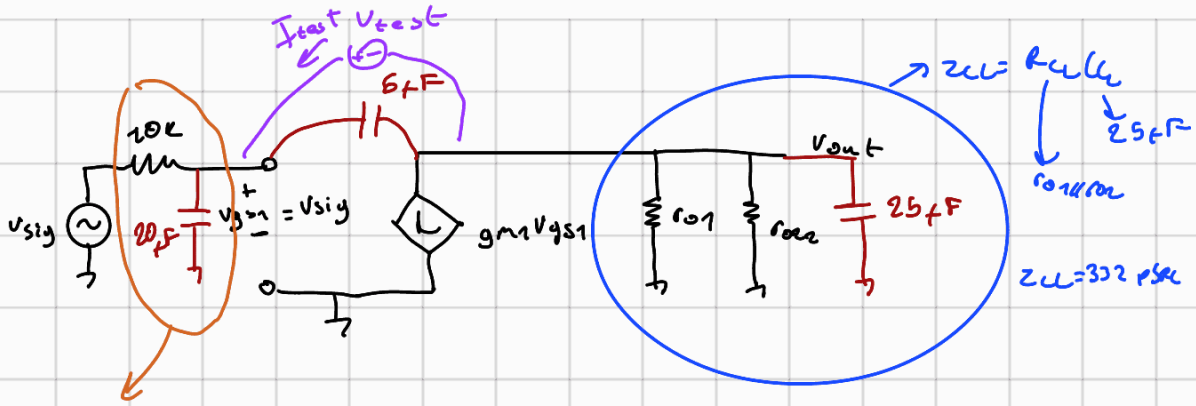
$$r_{o2} = \frac{10\,V/\mu m \times 0{,}4}{100\,\mu A} = 40\,k\Omega$$



$$A_m = -g_{m1}(r_{o1} \,//\, r_{o2})$$

$$= -1{,}25\,mS\,(20k \,//\, 40k)$$

$$A_m = -16{,}6\,[V/V]$$



$R_{G1} = R_{sig} = 10k\Omega$

$C_{G1} = C_{gs} = 20\ fF$

$C_{gs} = R_{G1} \cdot C_{G1} = 200\ psec$

$R_{gd} = R_{sig}(1 + g_m (r_{o1} \| r_{o2} \| R_L) + (r_{o1} \| r_{o2} \| R_L)$

$z_{gd} = R_{gd} C_{gd} = 950\ psec$

$z = z_{gd} + z_{gs} + z_{CL}$

$= 1482\ psec$

$f = \frac{1}{2\pi z} = 107{,}4\ MegHz$



$C_y = C_{gd}\left(1 + \frac{1}{g_m R_L}\right) = 5pF$

$C_x = (1 + g_m R_L') C_{gd}$

$C_x = 88\ pF$

$C_1 = 88 + 20 = 108\ pF$

$f_{p1} = \frac{1}{2\pi(10\Omega pF)} = 2{,}94\ MHz$

$C_2 = 30\ pF$

$f_{p2} = \frac{1}{2\pi(30Fx)\ 13{,}3k} = 399\ MHz$

$f_H = \sqrt{\frac{1}{\frac{1}{(f_{p1})^2} + \frac{1}{(f_{p2})^2}}}$

$f_H \cong 236{,}4\ Meg\,Hz$

